# Elizabeth Cady van Assendelft

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## PROFESSIONAL SUMMARY

Stanford Physics Ph.D. and quantum device researcher with 7+ years of experience in hardware development through its entire lifecycle, from initial circuit modeling and physical layout to final cryogenic characterization. Skilled in designing and implementing custom electronics and instrumentation to test superconducting devices. Effective science communicator recognized by national conference awards, peer-reviewed publications, and selection as a Stanford 3MT finalist.

## EDUCATION

**Ph.D. in Physics** | Stanford University, Stanford, CA *Defended Aug 2026; Degree conferral Dec 2026*

**B.S. in Physics (Intensive Track)** | Yale University, New Haven, CT *May 2019*

## CORE SKILLS/COMPETENCIES

- **Communication:** Science Translation, Technical Writing, Program Management, Mentorship
- **Hardware:** Superconducting Circuit Design, Quantum Device Fabrication, Cryogenic Systems, RF/Microwave Electronics, Rapid Prototyping, Vacuum Systems, PCB design
- **Software:** Python, MATLAB, COMSOL, SolidWorks, Fusion 360, Xic, Sonnet, FastHenry, WRSpice, Kicad
- **Operations:** Technical troubleshooting, Lab safety and EHS compliance, Process optimization, Standard Operating Procedures (SOPs), Confluence, Qualtrics

## PROFESSIONAL EXPERIENCE

**Graduate Research Assistant** | **Stanford University** *Sep 2019 – Dec 2026*

- **Quantum Hardware Development:** Designed, fabricated, and characterized Radio-frequency Quantum Upconverters (RQUs), a novel superconducting quantum sensor architecture.
- **Scientific Publishing & Collaboration:** Co-authored multiple peer-reviewed scientific papers in high-impact journals and presented technical findings at national conferences, including two invited talks.
- **Technical Communication & Recognition:** Selected as a finalist for the 2026 Stanford 3MT competition, and earned national recognition for research excellence, including the 2022 Graduate Instrumentation Research Award (Honorable Mention) from CPAD and the 2021 Student Presentation Award from the APS.
- **Lab Safety Oversight:** Served as Lab Safety Coordinator for a 15-person research facility, managing chemical inventories, hazardous waste, and Environmental Health & Safety (EHS) compliance; author Standard Operating Procedures (SOPs) for new instrumentation and train all incoming personnel.

**Program Coordinator** | **National Youth Science Camp** *Jun 2019 – Jul 2019*

- **Program Management:** Directed end-to-end operations for NYSCamp academic program, coordinating daily lecture series, directed studies, and special events led by visiting STEM experts.
- **Logistics and Communication:** Worked with Assistant Director to design, distribute, and execute comprehensive daily schedules and organized special events.

**Student Tech Coordinator** | **Student Technology Collaborative** *Jan 2016 – Jun 2019*

- **Technical Troubleshooting & Service Delivery:** Managed a high-volume, walk-in Technology Troubleshooting Office, providing real-time technical support and translating complex IT solutions for a diverse user base of undergraduate, graduate, and professional students.
- **Program Management:** Led and mentored a team of student technicians, optimizing the Silliman College help request queue and serving as the primary liaison between the tech collaborative and college administration.

**Undergraduate Research Assistant** | **Yale University** *Jan 2016 – May 2019*

- **Hardware Engineering, Instrumentation, & Upgrades:** Designed and implemented critical hardware upgrades for the HAYSTAC experiment, including mechanical control systems, complex dilution refrigerator interfaces, and implementation of a squeezed-state receiver that enabled a quantum-enhanced search for dark matter axions.
- **Prototype R&D:** Prototyped microfluidic test masses and built an atmospheric pressure optical trap to probe sub-millimeter gravitational interactions for the SIMPLE experiment.
- **Technical Communication & Recognition:** Earned 1st Place Poster at the 2018 APS Conference for Undergraduate Women in Physics and received 4 departmental and national awards for outstanding research contributions. Co-author on Nature paper.

## LEADERSHIP, TEACHING & COMMUNITY OUTREACH

**Science Policy Participant** | **Science Outside the Lab** *Jun 2026*

- Completed an intensive federal science policy immersion program, meeting with officials across the legislative and executive branches (EPA, GAO, OSTP, House Committee on Science, Space, and Technology) and advocacy organizations (ACLU, CSPO) to examine how technical evidence is translated for non-technical policymakers.

**Teaching Assistant** | **Stanford University, Yale University** *Jan 2019 – Mar 2025*

- Taught biweekly 80-minute lectures for freshman-level quantum and thermal mechanics, and upper-level undergraduate statistical mechanics. Managed twice-weekly lab sessions for graduate-level electronics laboratory.

**Graduate Student Coordinator** | **Stanford University** *May 2023 – Sep 2024*

- Ran 10-week summer research program for 50 undergraduate physics students for two consecutive years. Designed and ran professional development workshops and research presentations.

**STEM Outreach Volunteer** | **Yale University** *2018*

- Advocated for women in STEM by organizing outreach events for incoming students and teaching physics concepts to middle school students through Girls Science Investigations. Hosted and ran physics demonstration show for 200 middle school students.